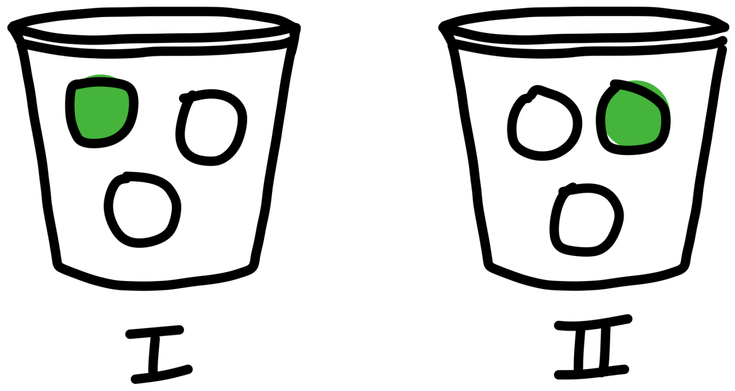
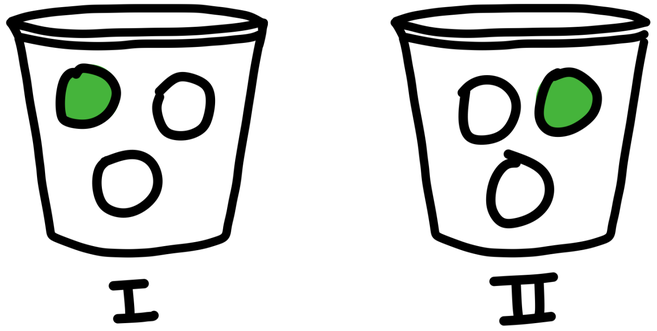
Example:

Two green balls and four white balls are put into two buckets in such a way that each bucket contains 3 balls.





At each time step, we randomly select one ball from each bucket and interchange them.

Let X_n = # green balls in bucket I after the nth exchange

We know, for example, that

$$P(X_4 = 1 \mid X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 0)$$

$$= P(X_4 = 1 \mid X_3 = 1) = \frac{5}{9}$$

Note that this probability does not actually depend on the particular time step, i.e. $P(X_{17} = 1 \mid X_{16} = 1) = \frac{5}{9}$

The chain is said to be

<u>time-homogeneous</u>.

(Default!)

We will organize the "one-step transition probabilities"

into a matrix.

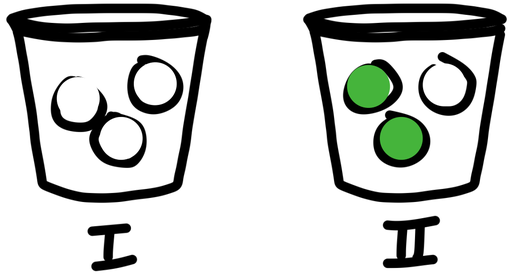
Transition Probability Matrix

"to" states

$$\mathbb{P} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} & & 0 \\ & 5/9 & \\ 0 & & \end{bmatrix} \end{array}$$

"from" states

$$P(X_1 = 1 \mid X_0 = 0) = ?$$



$$= P(\text{draw a G from II})$$

$$= 2/3$$

Transition Probability Matrix

$$\mathbb{P} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1/3 & 2/3 & 0 \\ 2/9 & 5/9 & 2/9 \\ 0 & 2/3 & 1/3 \end{bmatrix} \end{array}$$

<u>Note</u>: The rows of a transition probability matrix must sum to 1!

Notation: Let p_{ij} be the i-j th entry.

Then

$$p_{ij} = P(X_{n+1} = j \mid X_n = i)$$

and

$$\sum_j p_{ij} = 1.$$